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18 December 1967
Superseding
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10 October 1966

MILITARY SPECIFICATION

SWITCH, INERTIA, MARK 104 MOD 0

(For Torpedo Mk 46 Mods 0 and 1)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the requirements for procuring Inertia Switch Mk 104 Mod 0, referred to herein as the "switch." (See 6.1 for the description and intended use of the switch.)

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116	Preservations, Methods of

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

LD 547103	Switch, Inertia, and all documents listed thereon

FSC 1355

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(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order, the preproduction sample shall:

(a) Be manufactured using the procedures proposed for production.

(b) Consist of three switches.

(c) Be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.2 Basic requirements

3.2.1 Contractor-furnished data. Data required by this specification, or by applicable documents referenced in Section 2, need not be furnished unless specified in the contract or order (see 6.2).

3.2.2 Materials and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the switch shall be in accordance with the requirements specified herein and with the applicable documents listed in LD 547103.

3.2.2.1 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein, and which are necessary for the execution of this specification, shall be selected

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in accordance with Standard MIL-STD-143, except as provided herein. AN, JAN, or MS Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the AN, JAN, or MS Standard parts without alteration and provided the corresponding AN, JAN or MS part numbers are referenced on the parts list and, if practicable, on the contractor's drawing. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements specified herein.

3.2.3 Gages. The supplier shall provide whatever gages are necessary and adequate to ensure that the material and parts of the switch will meet dimensions and requirements shown on the applicable documents listed in LD 547103. When requested by the supplier, the procuring activity will furnish existing pertinent drawings of Naval final inspection gages for use only in the designing of the supplier's gages. This action does not relieve the supplier of the responsibility for designing and manufacturing the gages needed for the satisfactory fulfillment of the contract requirements.

3.2.4 Inspection and test equipment and facilities. The supplier shall furnish and maintain all necessary inspection and test equipment and facilities, and shall provide personnel for performing all acceptance tests specified herein. The test equipment shall be adequate to permit performance of the required acceptance tests.

3.3 Performance requirements and product characteristics

3.3.1 Sensitivity

3.3.1.1 Uniform acceleration sensitivity. The switch shall be capable of operation in accordance with Table I when subjected to uniform acceleration loads as specified in that table.

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Table I

Uniform Acceleration Sensitivity

Acceleration Direction * (See Figure 1)	Acceleration	
	Switch Open	Switch Closure **†
Switch Axis Perpendicular to Horizontal Plane with Leads Down	0 to 4.5 g's	4.5 to 7.0 g's
Switch Axis Parallel to Horizontal Plane with Leads toward Axis of Centrifuge	0 to 7.6 g's	7.6 to 12.6 g's
Switch Axis Parallel to Horizontal Plane with Leads Away from Centrifuge Axis	0 to 6.7 g's	6.7 to 11.4 g's

*See 6.3.3 for definition of Acceleration Direction.

**See 6.3.6 for definition of Switch Closure.

† Initial switch closure must occur between the values indicated and shall be maintained for any acceleration above the indicated higher level.

3.3.1.2 Shock sensitivity. The switch shall be capable of operation in accordance with Table II when subjected to a one-half sine wave shock pulse applied in any direction in the horizontal plane.

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Table II

Shock Sensitivity

Shock Load		Contact Response Required *	Closure Delay (Maximum) (ms)
Magnitude (g's)	Pulse Length (ms) ± 10%		
108	1	Close	9
63	1	Not Close	-
58	2	Close	11
26	2	Not Close	-
24	5	Close	12
7	5	Not Close	-
15	10	Close	15
5.5	10	Not Close	-
8.7	25	Close	24
4.6	25	Not Close	-

* Switch closure (see 6.3.6) shall be for not less than 50 microseconds duration.

3.3.1.3 Vibration sensitivity. The switch shall be capable of withstanding the vibration shown in Table III, in any axis, without closure.

Table III

Vibration Sensitivity

Double Amplitude (Inch, minimum)	Acceleration (g's, minimum)	Frequency (cps)
0.05	- -	5-20
-	1	20-31
0.02	- -	31-54
-	3	54-500

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3.3.2 Airtightness. Leakage of gas from the interior of the switch shall not exceed 1×10^{-5} cubic centimeters per second, when the external air pressure is reduced to 2.7 psia.

3.3.3 Contact resistance. The contact resistance of the switch, during closure, shall be less than 0.5 ohm, as measured between the two terminal wires of the switch.

3.3.4 Dielectric strength. The dielectric strength across the contact terminals of the switch in the normally open condition shall be not less than 10 megohms under an applied potential of 500 volts direct current (vdc).

3.3.5 Insulation resistance. The insulation resistance of both contacts (shorted) to case shall be not less than 100 megohms under an applied potential of 500 volts direct current (vdc).

3.4 Environmental requirements. Unless otherwise specified, the switch shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed to the environments specified in this section.

3.4.1 Vibration

3.4.1.1 Ready nonoperating and ready condition. The switch shall be capable of withstanding, in the ready condition, the vibrations defined in Table IV. The switch shall be vibrated for 180 minutes in each horizontal axis and in two mutally perpendicular transverse axes at any temperature within the range of -45 to +140°F.

3.4.1.2 Nonoperating condition. The switch, packed and packaged in accordance with Section 5, shall be capable of withstanding the vibration defined in Table IV.

TABLE IV

Vibration Schedule

Condition (see 6.3.5)	Double Amplitude (in.) Minimum	Acceleration (g's) Minimum	Frequency Range (cps)
Nonoperating	--	1.0	5-60
Ready Nonoperating and Ready	0.2 --	-- 3.0	5-17 17-500

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3.4.2 Shock (see 6.3.2). The switch shall be capable of withstanding the shock specified in Table V.

Table V

Shock Requirements

Condition (see 6.3.5)	Peak Magnitude (g's) ± 10%	Minimum Duration (ms)	Acceleration Direction *
Nonoperating	60	8.0 to 10.0	Any
Ready	45	18.0 to 20.0	Any in horizontal plane
	85	0.5 to 1.5	Any
	300	0.5 to 1.5	Any in horizontal plane

* See 6.3.3 for definition of Acceleration Direction.

3.4.3 Acceleration (see 6.3.3). The switch, in the ready condition (see 6.3.5), shall be capable of withstanding an acceleration of 35 g's in the horizontal plane.

3.4.4 Temperature

3.4.4.1 Temperature, nonoperating. The switch shall be capable of conforming to the requirements of this specification after being at any temperature within the range of -45 to 140 degrees F while in the nonoperating condition (see 6.3.5.1), and while at any temperature within the range of 0 to 140 degrees F while in the ready nonoperating and ready conditions (see 6.3.5.2 and 6.3.5.3).

3.4.4.2 Temperature, operating. The switch shall be capable of conforming to the requirements of this specification while at any temperature between 0 and 140 degrees F.

3.4.5 Humidity (see Section 6.3.4). The switch shall be capable of withstanding exposure to 95 percent humidity at 110 ± 5°F for 8 hours without degradation of functional capabilities.

3.5 Storage life

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3.5.1 Storage life, ready. The switch, in the ready condition (6.3.5.3), shall be capable of meeting the performance requirements specified herein after one year of unattended storage.

3.5.2 Storage life, nonoperating. The switch shall be capable of meeting the performance requirements specified herein subsequent to storage, in a packaged and preserved condition as specified in 5.1.1, for a period of five years.

3.6 Identification. The switch shall be permanently identified in accordance with the applicable documents listed in LD 547103.

3.7 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the switch shall be classified as follows:

(a) Preproduction inspection

(1) Preproduction sample (see 4.3.1)

(2) Preproduction tests (see 4.5)

(b) Periodic production inspection

(1) Periodic production sampling (see 4.3.2)

(2) Periodic production tests (see 4.6)

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(c) Quality conformance inspection

(1) Quality conformance sampling (see 4.3.3)

(2) Quality conformance tests and examinations (see 4.7)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of three switches, manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.5 at a facility designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Periodic production sample. Periodic production samples shall be selected from units that have satisfactorily passed the quality conformance inspections of Section 4.7. These samples shall be subjected to the tests of Section 4.6 at an activity designated by the procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1.

4.3.2.1 When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

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(d) Submit to the procuring activity for approval, the proposed retest program to verify that the corrective action eliminated the defect.

Return to Step 1 and proceed as indicated when inspection is resumed.

4.3.3 Quality conformance sampling. The tests and examinations detailed in 4.7 shall be conducted on each switch to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units shall be rejected.

4.4 Test conditions. Unless otherwise specified, the switch shall be subjected to tests under the following conditions:

(a) Temperature: Room ambient (60 to 95 degrees F).

(b) Humidity (relative): Room ambient, up to 95 percent.

(c) Altitude: Normal ground atmospheric pressure.

4.5 Preproduction tests. The switches within the preproduction sample shall be subjected to the following preproduction tests (4.5.2 through 4.5.4) in the order listed. Each switch shall be subjected to and satisfactorily pass the quality conformance tests and examinations (4.7) prior to the preproduction tests. Certified inspection data covering the results of preproduction tests shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Date of tests and names of persons conducting tests.

(b) Run or test plan used.

(c) Serial number of switch.

(d) Test results of preproduction tests (4.5.2 through 4.5.4).

4.5.1 Preproduction test equipment. The following items of test equipment are required to perform the preproduction tests specified herein.

(a) Centrifuge or rate table, capable of imparting accelerations to a 0.1 pound item of up to 13 g's in the horizontal plane.

(b) Oscilloscope, 5-inch diameter tube, with less than 2.0 percent ripple, and a rise time of 5 microseconds or less. Associated with the oscilloscope is the test circuit of Figure 3.

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(c) Megohmmeter, with an integral source of 500 ± 50 vdc and a midscale reading in the order of 10 megohms, with an error of less than 10 percent.

(d) Vibration test equipment, capable of testing the switch at the frequencies and amplitudes required by Tables III and IV.

(e) Shock test equipment, capable of testing the switch at the magnitudes and durations required by Tables II and V.

(f) Acceleration test equipment, capable of subjecting the switch to an acceleration of 35 g's in the horizontal plane.

(g) Temperature controlled chamber, capable of subjecting the switch to the temperature cycle of 4.5.2.4.

(h) Power source capable of supplying 12 volts dc.

(i) Power source capable of supplying 100 volts dc.

4.5.2 Preproduction tests. The switches within the preproduction sample (identified as switches 1, 2, and 3) shall be subjected to the tests of 4.5.2.1 through 4.5.4. Acceptance shall be based on the absence of any defective units within the group.

4.5.2.1 Vibration, ready condition. The switch, in the ready condition shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The exploder shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table VI. If more than 4 resonances are encountered in any axis, the 4 most severe resonances shall be chosen for the test. The resonant cycling period for each axis shall be divided into 3 equal parts, which are conducted at -45°F ±5°F, room temperature, and +140°F ±5°F respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table IV.

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Table VI

Resonance & Cycling Schedule

No. of Resonance	Time at Resonance at Each Temperature in Each Axis, And at Each Resonance (Min)	Cycling Time at Each Temperature in Each Axis (Min)
0	0	60
1	10	50
2	10	40
3	10	30
4	10	20

4.5.2.2 Shock. Mount the switch as shown in Figure 1 and secure it to shock test equipment (see 4.5.1 (e)) so that a minimum of 90 percent of the table input of the fixture will be applied to the switch under test. Subject the switch, at room temperature, to the shock listed in Table VII, using the procedure specified in Shock Test Procedure IV of MIL-E-5272 except as follows: Shock magnitude and duration shall be measured by an accelerometer mounted on the input table of the fixture to indicate applied shock. Test equipment is not restricted to that specified in MIL-E-5272.

Table VII

Shock Schedule

Peak Magnitude (g's) ± 10%	Minimum Duration (ms)	Impact Direction (see Fig. 1)	Total Number of Impacts
300	0.5 - 1.5	Horizontal Plane *	3
85	0.5 - 1.5	Vertical Axis	6
60	8.0 - 10.0	Any	9
45	18.0 - 20.0	Horizontal Plane *	3

* Apply shocks in three axes of the horizontal plane, each separated by 120° (one at 0°, one at 120°, and one at 240°).

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4.5.2.3 Acceleration. Mount the switch as shown in Figure 1, attach it to acceleration equipment (4.5.1 (f)), and subject it to the acceleration magnitude (± 10%) and direction required by 3.4.3, for one minute at room temperature.

4.5.2.4 Temperature and humidity; ready nonoperating condition. The switch, in the ready nonoperating condition, shall be subjected to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45°F ±5°F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45°F ±5°F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours.) The specimen shall then be subjected to an environmental temperature of +140°F ±5°F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110°F ±5°F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Immediately thereafter, the switch shall be operated in accordance with Section 4.7 (except do not conduct 4.7.2). Prior to operation, the switch may be wiped dry. Drying by an air blast or oven is not permitted.

4.5.2.5 Operating temperature

4.5.2.5.1 Low temperature, operating. Before testing, the switch shall be installed in its test chamber in the ready condition and conditioned at a temperature of 0°F ±5°F for a period of 1 hour. The switch shall then be operated in accordance with and meet the requirements of Section 4.7 (except do not conduct 4.7.2) at this temperature.

4.5.2.5.2 High temperature, operating. Before testing, the switch shall be installed in its test chamber in the ready condition and conditioned at a temperature of +140°F ±5°F for a period of 1 hour. The switch shall then be operated in accordance with and meet the requirements of Section 4.7 (except do not conduct 4.7.2) at an environmental temperature of +140°F ±5°F.

4.5.2.6 Vibration, nonoperating. The switch, packed and packaged in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing minimum visual movement of the pack within the frequency range of 5 to 60 cps. The container shall be placed on the vibration platform and fenced on all sides so that the pack cannot fall off the platform. Starting at 5 cps the frequency shall be gradually increased until the pack just begins to leave the platform. The specimen shall then be vibrated for 1 hour.

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4.5.3 Vibration sensitivity. Mount the switch as shown in Figure 1 and, with leads connected as shown in Figure 4, conduct a resonance survey in the horizontal plane for 15 minutes minimum between 5 and 500 cycles per second (cps) (in accordance with 3.3.1.3 and Table III) and note all resonant frequencies. Connect the switch leads to equipment as shown in Figure 2. Vibrate the switch at each resonant frequency for 15 seconds minimum. The acceleration or double amplitude value shall be as shown in Table III. A closure (6.3.6) during this test constitutes a failure.

4.5.4 Shock sensitivity. Mount the switch on a shock fixture (see 4.5.1 (e)), so that shock loads will be applied parallel to the horizontal plane (see Figure 1). Connect the switch leads to equipment as shown in Figure 2. Apply shock loads in accordance with Table II (3.3.1.2) and monitor switch closure. Switch sensitivity not in accordance with Table II constitutes a failure.

4.6 Periodic production tests. The periodic production sample selected in accordance with 4.3.2, shall be subjected to the following tests (4.6.2). Certified inspection data covering the tests results shall be forwarded to the Naval Ordnance Systems Command. Certified inspection data shall consist of the following:

(a) Item 4.5 (a) through 4.5 (c).

(b) Test results of periodic production tests (Table VIII).

4.6.1 Periodic production test equipment. The equipment required for periodic production testing shall be the same as that specified in 4.5.1.

4.6.2 Periodic production test schedule. Periodic production tests shall be conducted in accordance with Table VIII.

Table VIII

Test	Sample & Test Sequence 1, 4, etc.	2, 5, etc.	3, 6, etc.
Vibration (4.5.2.1)	1*		1*
Acceleration (4.5.2.3)		1*	2*
Shock (4.5.2.2)	2*		3*
Temperature & Humidity (ready nonoperating)(4.5.2.4)		2	4
Hi-Temperature (operating)(4.5.2.5.2)	3		5
Lo-Temperature (operating)(4.5.2.5.1)		3	6
Vibration Sensitivity (4.5.3)		4	7
Shock Sensitivity (4.5.4)	4		8
Final Quality Conformance (4.7, but excluding 4.7.2)	5	5	9

* Visual check after testing.

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4.7 Quality conformance tests and examinations. The following tests and examinations (4.7.2 through 4.7.5) shall be conducted on each switch to be delivered. Failure to pass any test or examination specified herein shall be considered cause for rejection of the unit.

4.7.1 Quality conformance test equipment. The following items of test equipment are required to perform the acceptance tests specified herein:

(a) Centrifuge or rate table (see 4.5.1 (a)).

(b) Oscilloscope (see 4.5.1 (b)).

(c) Megohmmeter (see 4.5.1 (c)).

(d) DC power source (see 4.5.1 (i)).

(e) DC power source (see 4.5.1 (j)).

(f) Leak detector, capable of measuring leakage of helium at a rate of 1 x 10^{-5} atmospheric cc's per second and at a pressure of 2.7 psia.

(g) A pressure-vacuum chamber capable of pressures from 60 psig to 2.7 psia.

4.7.2 Sensitivity and contact resistance. Test equipment shall be set up for this test in accordance with Figure 3 and 4.7.1.

(a) Turn on the oscilloscope, 12 volt dc power supply, and 100 volt dc power supply. Short circuit TB2 to TB5 and operate the centrifuge at an average test rpm (80 to 100).

(b) Adjust oscilloscope vertical channel so that the trace is at the upper portion of the screen. Set the oscilloscope sweep rate at 0.1 millisecond per centimeter.

(c) Stop the centrifuge and remove the short circuit between TB2 and TB5. Set the calibration switch at Position 1, and adjust R1 to obtain the same reading on the oscilloscope as in 4.7.2(b). Return the calibration switch to Position 3.

(d) Place a 0.5 ohm plus or minus 1 percent resistor across TB2 and TB5 and operate the centrifuge at an average test rpm (80 to 100).

(e) Adjust the oscilloscope sensitivity so that the trace is 5 centimeters minimum below that in 4.7.2(b).

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(f) Stop the centrifuge and remove the 0.5 ohm resistor from between TB2 and TB5. Set the calibration switch at Position 2, and adjust R2 to obtain the same reading on the oscilloscope as in 4.7.2(e). Return the calibration switch to Position 3.

(g) Mount the switch on the centrifuge (with electrical leads down) so that the horizontal plane of the switch is parallel to the plane of centrifugal rotation and in a manner that allows rotation of the switch about the switch axis during rotation of the centrifuge (see Figure 5).

(h) When a momentary downward excursion is observed on the oscilloscope screen, note the reading on the centrifuge rpm indicator.

(i) Inertia switch contact resistance not as specified in 3.3.3 and sensitivity of the inertia switch not as specified in Table I constitute a failure.

(j) Slowly increase the centrifuge rpm to obtain a maximum acceleration on the inertia switch of 7 g's and observe that steady closure is achieved with a contact resistance of less than 0.5 ohms.

(k) Maintain the 7 "g" acceleration and rotate the inertia switch about the inertia switch axis through 360°. The minimum time to complete 360 degrees of continuous rotation is 4 minutes.

(l) When a resistance greater than 0.5 ohm is observed, record the rotation angle. Also record the span of high resistance in degrees of switch rotation. Loss of switch closure (switch closure not in accordance with Table I), which is verified by stopping the centrifuge and returning to the same "g" load, shall constitute a failure.

(m) Continue inertia switch rotation until a continuous scan of 360 degrees has been achieved.

(n) Repeat steps (g) through (m), except that the "g" level shall be 4.5 g's maximum and the switch shall be monitored for no closure.

(o) Mount the inertia switch so that its axis is parallel to the plane of centrifuge rotation and the inertia switch leads are toward the pivotal axis of the centrifuge.

(p) Start the centrifuge and slowly increase the acceleration. When switch closure presentation is observed, make a notation of the resistance and note the reading on the centrifuge rpm indicator.

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(q) Inertia switch contact resistance not as specified in 3.3.3 and sensitivity of the inertia switch not as specified in Table I constitute a failure.

(r) Mount the inertia switch so that its axis is parallel to the plane of centrifuge rotation and the inertia switch leads are away from the pivotal axis of the centrifuge.

(s) Start the centrifuge and slowly increase the acceleration. When switch closure presentation is observed, make a notation of the resistance and note the reading on the centrifuge rpm indicator.

(t) Inertia switch contact resistance not as specified in 3.3.3 and sensitivity of the inertia switch not as specified in Table I constitute a failure.

4.7.3 Dielectric strength. Apply the test probes of a megohmmeter (4.7.1(c)) to the contact terminals of the switch when it is in the normally open condition. With a potential of 500 vdc across the contact terminals, a dielectric strength of less than 10 megohms (see 3.3.4) constitutes a failure.

4.7.4 Insulation resistance. Apply the test probes of a megohmmeter (4.7.1(c)) to the contacts (shorted) and case of the switch. When a potential of 500 vdc is applied, an insulation resistance of less than 100 megohms (see 3.3.5) constitutes a failure.

4.7.5 Airtightness. This test shall be performed after the switch is sealed. The switch shall be placed in the pressure vacuum chamber and pressurized at 60 psig in a helium atmosphere for 4 hours. The helium atmosphere shall then be vented and pressure reduced to 2.7 psia or less. The leak detector shall be employed to test for helium leakage for a period of no less than 4 minutes, in accordance with the requirements of 3.3.2.

4.7.6 Examination. The switch shall be examined for conformance to the requirements specified herein and on the applicable NOSC drawings. The presence of any one of the following defects shall constitute a failure:

(a) Switch not complete; component parts missing.

(b) Component parts not new; evidence of reclaimed, used or salvaged components.

(c) Inadequate solder joints.

(d) Evidence of use of unauthorized material and repairs.

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(e) Switch configuration extends beyond drawing limits.

(f) Inadequate external switch insulation.

(g) Dimensions do not comply with drawing requirements.

(h) Surface roughness does not comply with drawing requirements.

(i) Lead wire gage incorrect; lead wires short.

(j) Identification plate and markings not attached; incorrectly located; data illegible or incomplete.

(k) Preservation, packaging, or packing not as specified in Section 5.

5. PREPARATION FOR DELIVERY

(a) Unless otherwise specified (see 5(b)), the switch shall be packaged and packed as an integral component of Exploder Mechanism Mk 20 Mod 0, in the containers specified for the exploder mechanism, as set forth in the applicable NOSC drawings.

(b) When procured as spare or replacement parts, or otherwise procured not as an integral component of the exploder mechanism, the switches shall be preserved, packaged, and packed in accordance with the levels established below, as specified by the procuring activity (see 6.2).

5.1 Preservation and packaging. Preservation and packaging of the switch shall be Level A or C as specified (see 6.2).

5.1.1 Level A. Cleaning shall be in accordance with Process C-1 of MIL-P-116. Drying, if required, shall be in accordance with Process D1 or D4 of MIL-P-116. Preservation and packaging shall be in accordance with Method I-A of MIL-P-116.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging of the switch shall be in accordance with the supplier's commercial practices, when such practices have been approved by the procuring activity.

5.2 Packing. Packing of the switch shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. The switch, preserved and packaged in accordance with 5.1.1, shall be packed as specified by the procuring activity (see 6.2).

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5.2.2 Level B. Not applicable.

5.2.3 Level C. The switch, preserved and packaged in accordance with 5.1.3, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Description and intended use

6.1.1 Description. Switch Mk 104 Mod 0 is an inertia operated electric switch with normally open contacts. The switch consists of a ball weight unit which is retained in position by a spring force, and surrounded by a metal shell. The switch is sealed in a plastic housing with two electrical leads extended from it. One lead is electrically connected to the ball unit and the other connected to the shell. The switch circuit is completed when the ball makes contact with the metal shell.

6.1.2 Intended use. The switch is intended for use as the firing switch in an exploder mechanism.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation, packaging, and packing.

(c) Serialization requirements.

(d) Assigned activity for preproduction and periodic production tests.

(e) Data required.

(f) All additional samples required shall be provided and inspected at the expense of the supplier when such additional samples are required because of failure of any switch(es) to pass the prescribed periodic production inspection.

(g) Any exceptions to this specification.

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(h) That certified inspection data covering the results or preproduction and periodic production inspections shall be forwarded to the Naval Ordnance Systems Command when testing is accomplished at other than a Government activity.

6.3 Definitions

6.3.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (cps).

6.3.2 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in units of gravitational acceleration (g's), duration in units of time, milliseconds (ms), and shock pulse approximating a one-half sine wave. The time duration shall be measured between points where two lines, tangent at maximum slope, intersect the "zero" g line.

6.3.3 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in multiples of gravitational units (g's). Acceleration direction is opposite to inertia load and points toward center of rotary table.

6.3.4 Humidity. The relative humidity of the environment to which the switch is exposed, expressed as a percent.

6.3.5 Switch condition

6.3.5.1 Nonoperating. The condition of the uninstalled switch when stored, transported, or handled.

6.3.5.2 Ready, nonoperating. The condition of the switch when assembled with the end item, transported, and handled.

6.3.5.3 Ready. The condition of the switch when fully assembled with the torpedo and installed on the launching mechanism prior to operating.

6.3.5.4 Operating. The condition of the switch when operating.

6.3.6 Switch closure. Switch closure is characterized by electrical continuity through the switch with resistance less than 0.5 ohm. Possible hash or wavy line presentation is permissible, depending upon equipment and circuit characteristics.

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6.3.7 Defective units. A unit shall be considered defective if it does not meet all examinations or tests specified, or if deviations are found through visual examination after environmental exposure.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

6.4.1 Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

6.4.2 Level B. Not applicable.

6.4.3 Level C. This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installation for immediate use at the first receiving activity.

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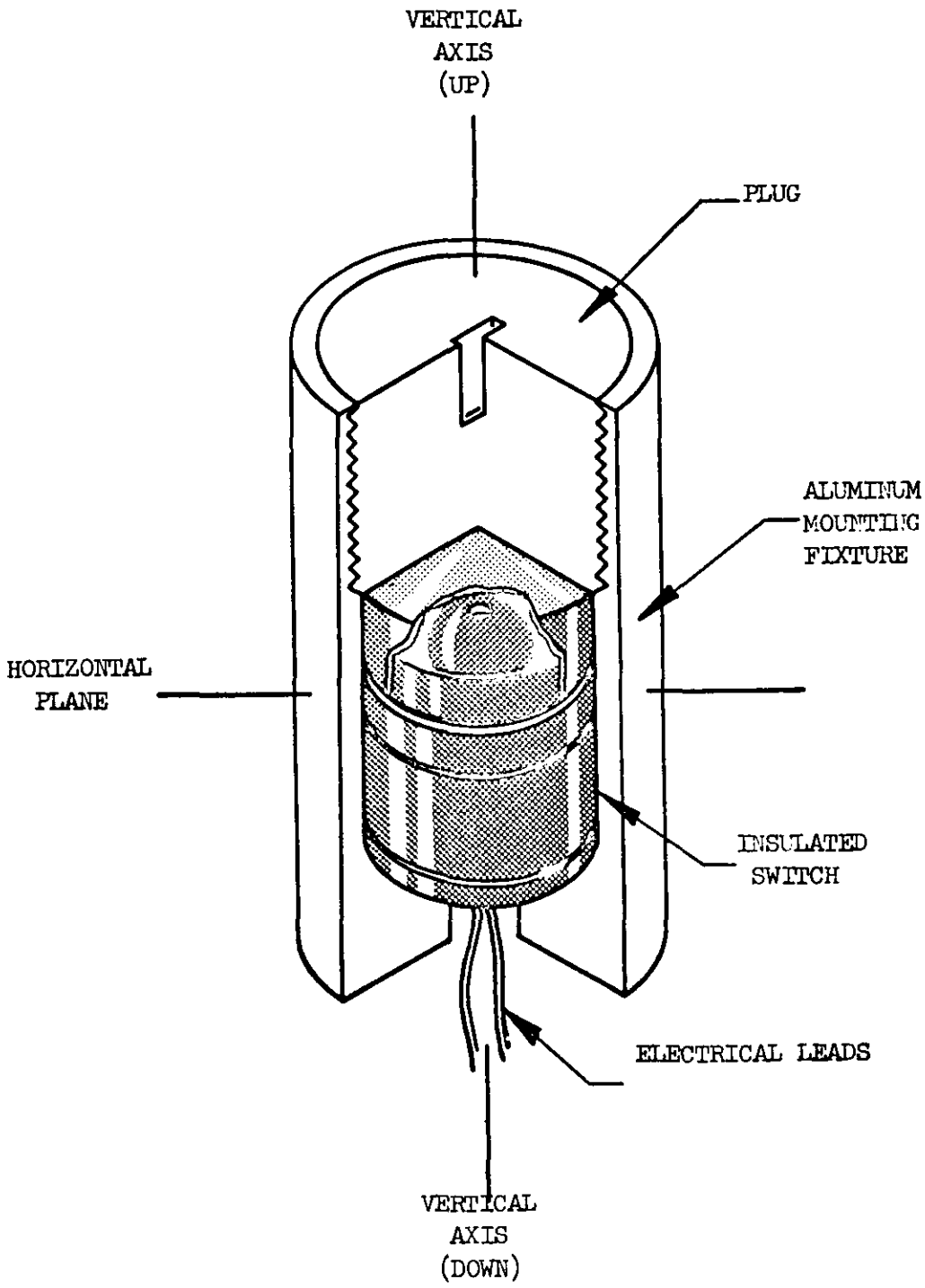


FIGURE 1. MOUNTING FOR ENVIRONMENTAL TESTING

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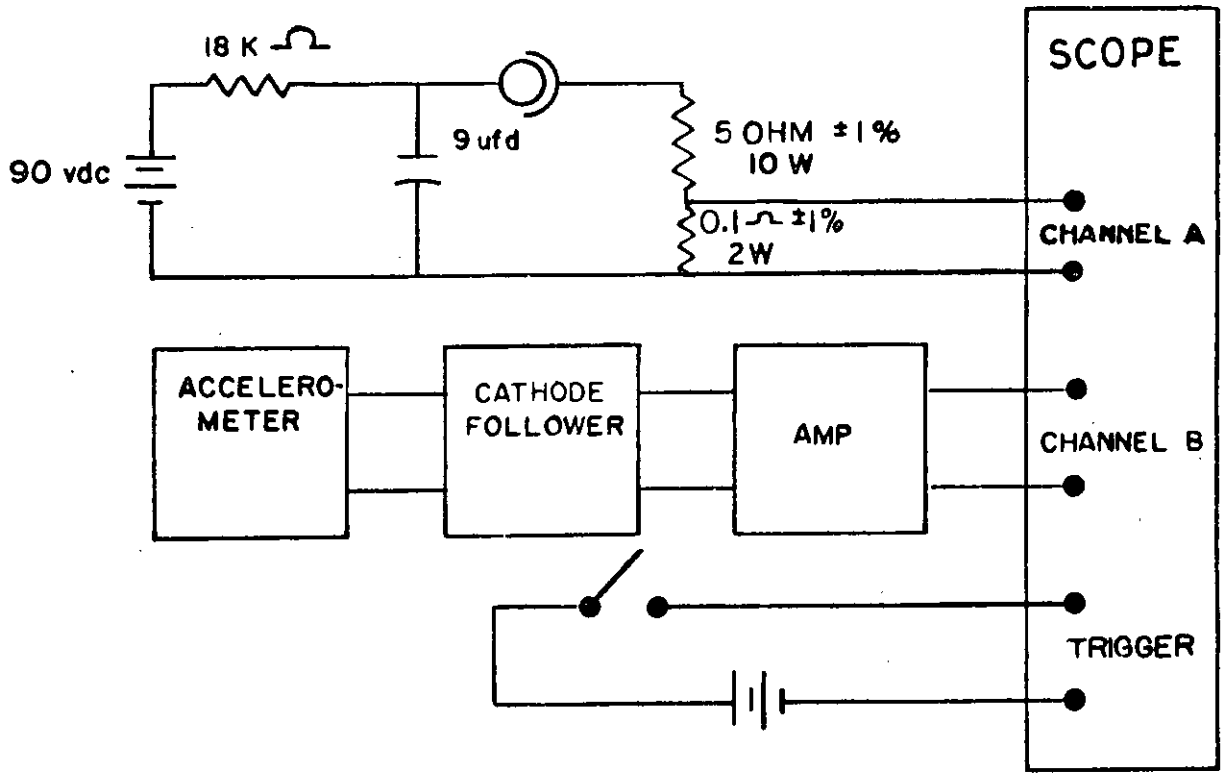


FIGURE 2. SHOCK SENSITIVITY TEST CIRCUIT

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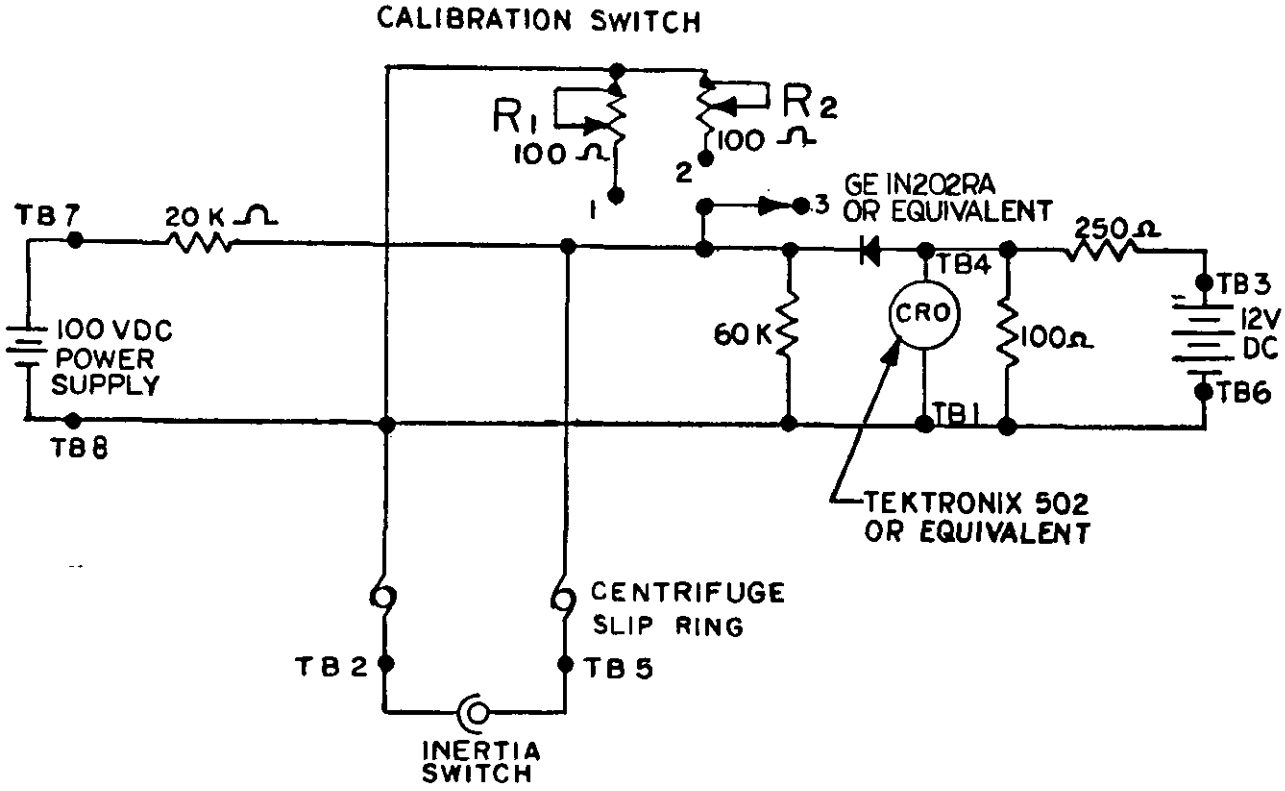


FIGURE 3. SENSITIVITY AND CONTACT RESISTANCE TEST CIRCUIT (75 VOLT OPEN SWITCH POTENTIAL)

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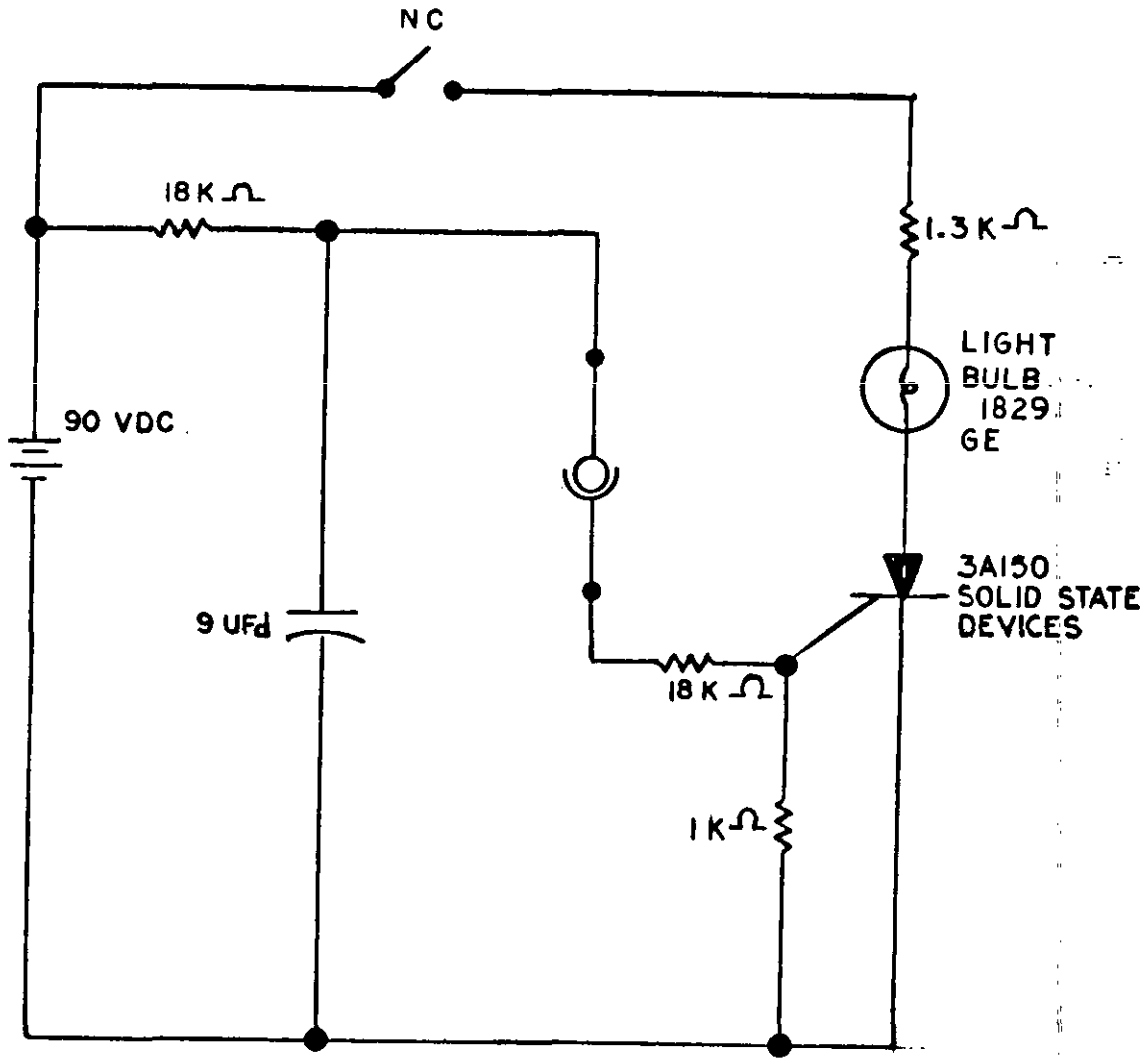


FIGURE 4. VIBRATION SENSITIVITY TEST CIRCUIT

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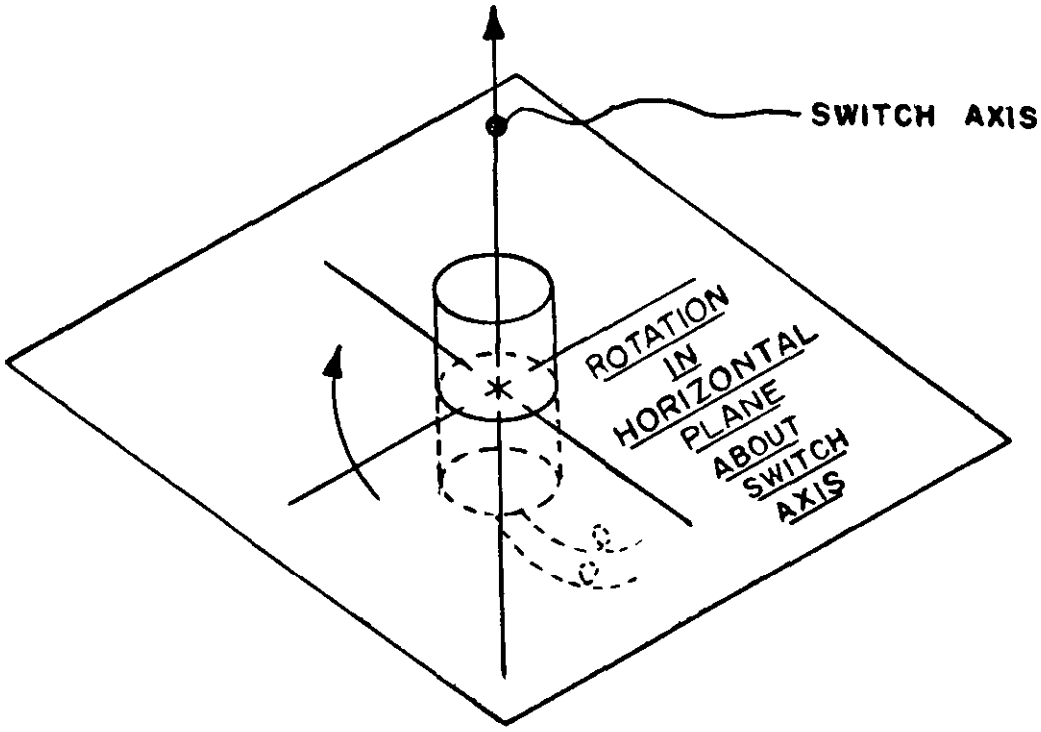


FIGURE 5. ROTATION IN HORIZONTAL PLANE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-S-81031B(OS) SWITCH, INERTIA MK 104 MOD 0(For Torpedo Mk 46 Mod 0 and 1)

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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